



Computer Networks

Week # 04

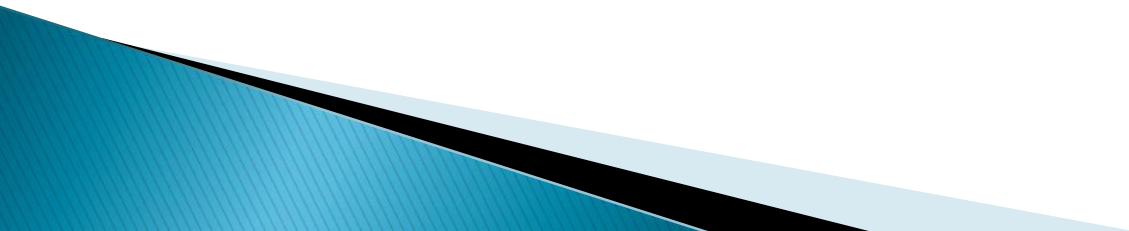
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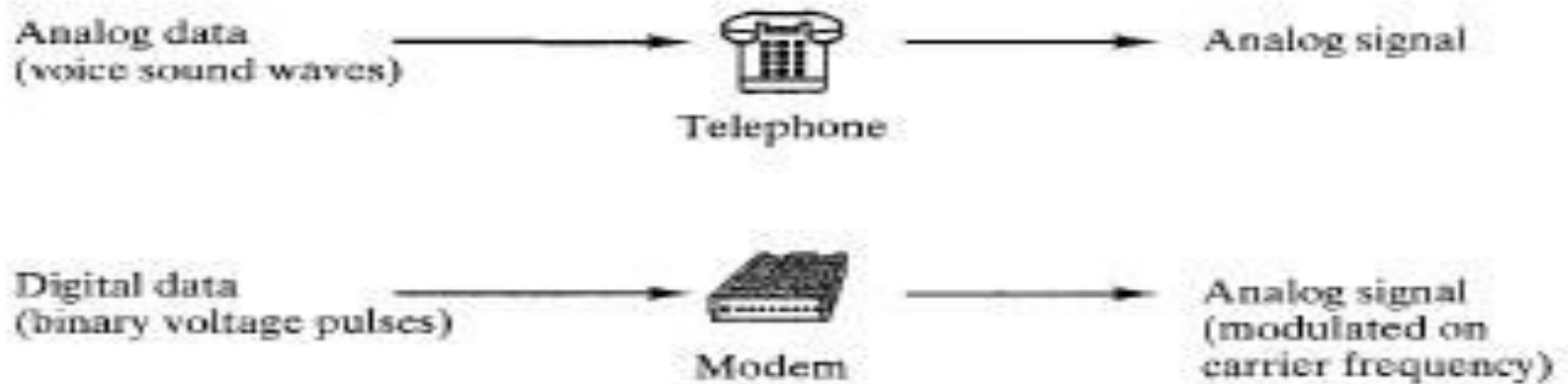
URL- <https://sites.google.com/view/nadeemuolcscn/home/course-material>

ACKNOWLEDGMENTS

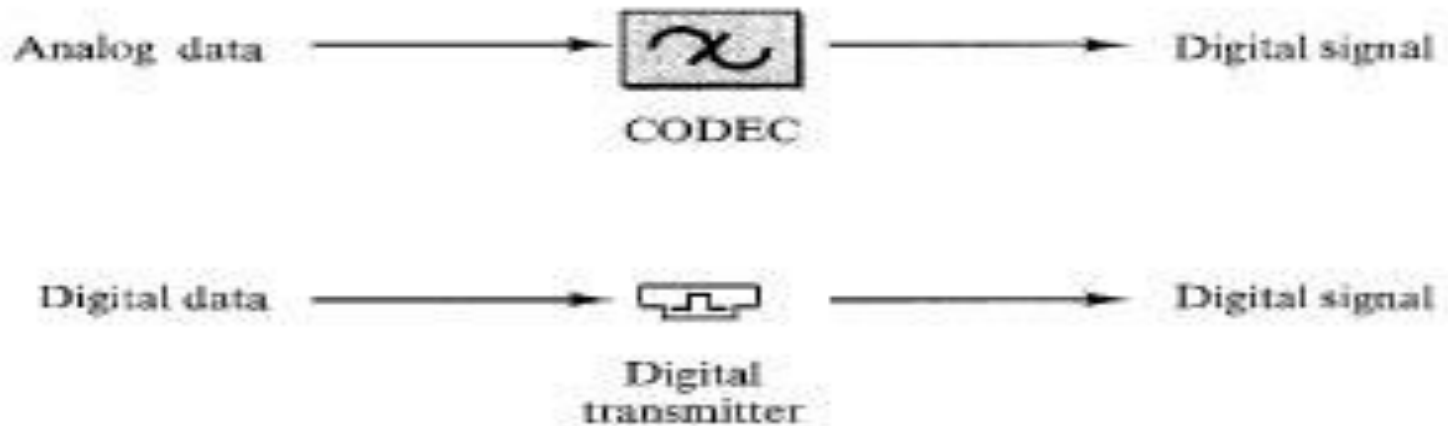
These lecture slides contain material from slides prepared by Behrouz Forouzan for his book Data Communication and Networking (4th/ 5th edition).



Analog/Digital Conversions

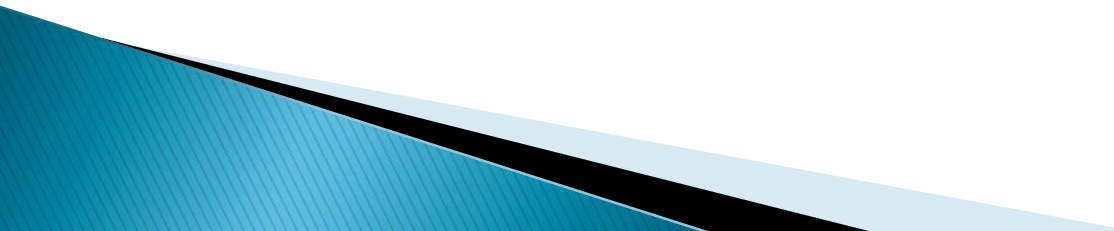


(a) Analog signals: represent data with continuously varying electromagnetic wave



(b) Digital signals: represent data with sequence of voltage pulses

Digital-To-Digital Conversion

- Representation of digital data by using digital signals
 - Techniques for digital-to-digital conversion
 - ❖ Line coding
 - ❖ Block coding
 - ❖ Scrambling
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Digital-To-Digital Conversion

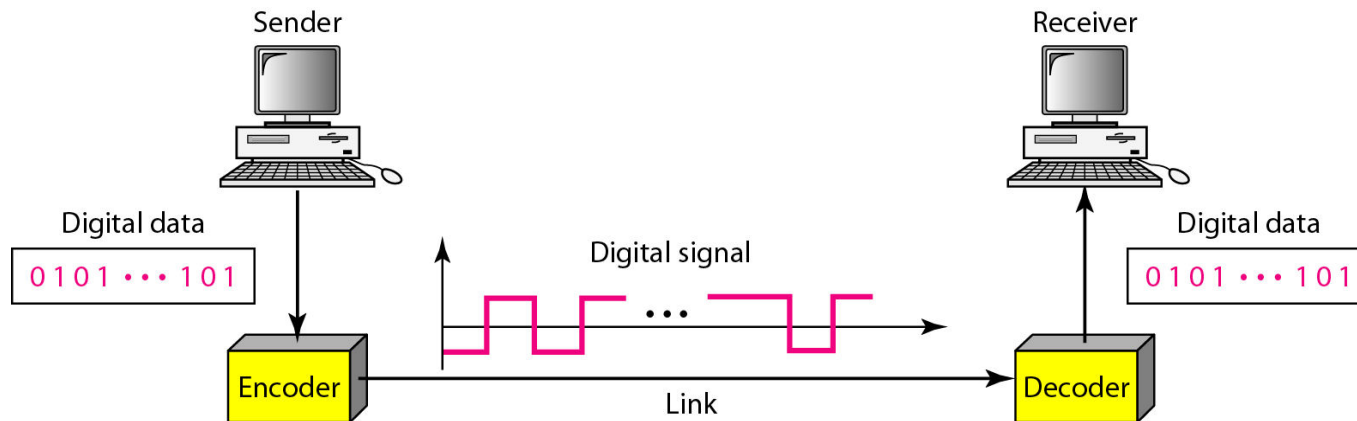
Line Coding: the process of converting digital data to digital signals

- **Assumption:** Data is stored in computer memory as sequence of bits
- Line coding converts sequence of bits to digital signals
- For example when we transmit data from computer to the printer, both original and transmitted data have to be digital
- Encoding a digital signal is where 1's and 0's generated by the computer are translated into voltage pulses that can be propagated over the wire

Digital-To-Digital Conversion

Line Coding

- **At sender:** encoding of digital data into digital signal
- **At receiver:** recreation of digital data by decoding the digital signal



Digital-To-Digital Conversion

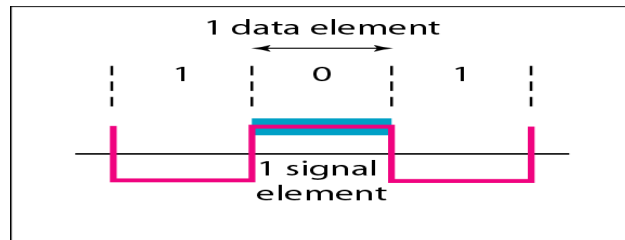
Characteristics of Line Coding Schemes

- **Signal element versus data element**
- **A data element** is the smallest entity (bit) that can represent a piece of information
 - ❖ In data communications, our goal is to send data elements which are what we need to send
 - ❖ Data elements are being carried
- **A signal element** is the shortest unit (time wise) of a digital signal
 - ❖ In digital data communications, a signal element carries data elements (carriers)
 - ❖ So signals elements are what we can send
 - ❖ **Ratio** of data elements to signals elements

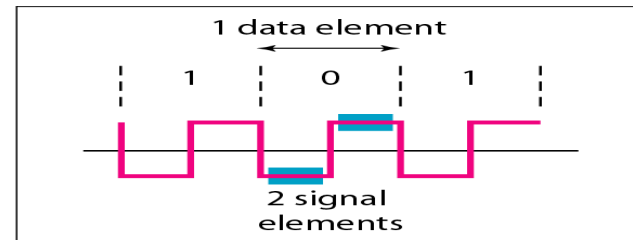
Digital-To-Digital Conversion

Characteristics of Line Coding Schemes

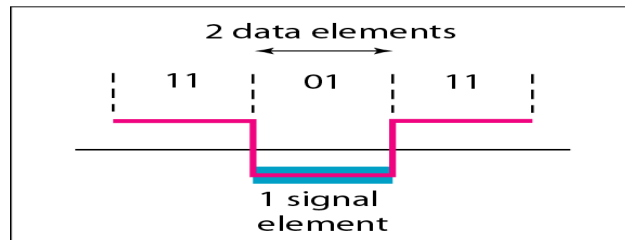
- **Signal element versus data element**
- In the simplest case, there is a one-to-one correspondence between bits and signal elements



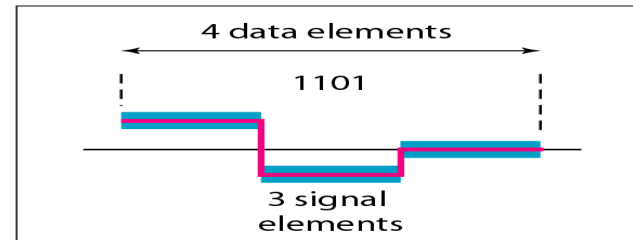
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Digital-To-Digital Conversion

Characteristics of Line Coding Schemes

- **Data rate versus signal rate**
- The **data rate** (bit rate) is the number of data elements (bits) sent in **1 second**
 - ❖ The unit is bits per second (bps)
- The **signal rate** (baud rate, modulation rate or pulse rate) is the number of signal elements sent in **1 second**.
 - ❖ The unit is the baud
- **Data communications goal:** increase the data rate (increasing speed of transmission) and decrease the signal rate (decreasing bandwidth requirement)

Digital-To-Digital Conversion

Characteristics of Line Coding Schemes

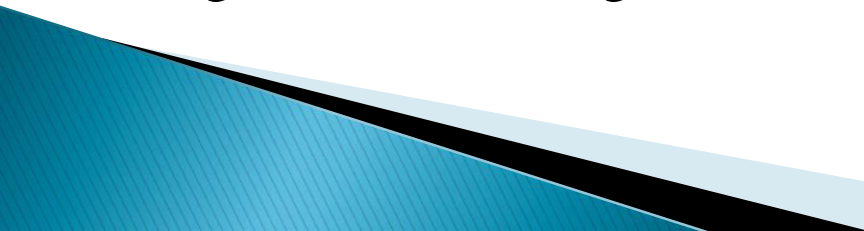
Bandwidth

- Although the actual bandwidth of a digital signal is infinite, the effective bandwidth is finite
- the baud rate, not the bit rate, determines the required bandwidth for a digital signal

Digital-To-Digital Conversion

Characteristics of Line Coding Schemes

Baseline wandering

- In decoding a digital signal
 - ❖ the receiver calculates a running average of the received signal power. This average is called the *baseline*.
 - To determine the value of the data element
 - ❖ *The incoming* signal power is evaluated against this baseline
 - A long string of 0s or 1s can cause a drift in the baseline (**baseline wandering**) and make it difficult for the receiver to decode correctly
 - A good line coding scheme needs to prevent baseline wandering
- 

Digital-To-Digital Conversion

Characteristics of Line Coding Schemes

DC component

- When the **voltage level** in a digital signal is **constant** for a while, the **spectrum** creates very **low frequencies** (results of Fourier analysis)
- These frequencies around zero, called **DC (direct-current) components**, *present problems*
 - ❖ *for a system that cannot pass low frequencies.*
 - ❖ For example, a telephone line cannot pass frequencies below 200 Hz
- So DC component means 0/1 parity that can cause baseline wandering
- we need a scheme with no DC component for such a system

Digital-To-Digital Conversion

Characteristics of Line Coding Schemes

Self synchronization

- To correctly interpret the signals received from the sender,
 - ❖ the receiver's bit intervals must correspond exactly to the sender's bit intervals.
- If the receiver clock is faster or slower
 - ❖ the bit intervals are not matched and the receiver might misinterpret the signals
- Consider a situation where the receiver has a shorter bit duration
 - ❖ Let the sender sends 10110001, while the receiver receives 110111000011

Digital-To-Digital Conversion

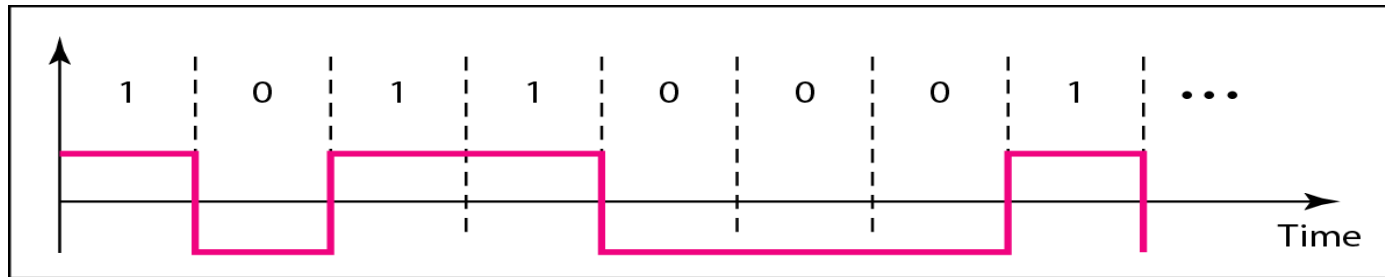
Characteristics of Line Coding Schemes

Self synchronization

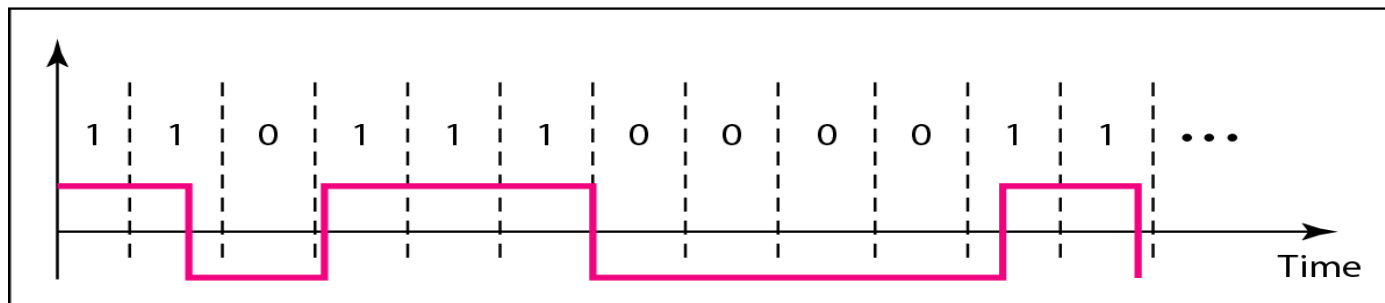
- A self-synchronizing digital signal includes timing information in the data being transmitted
 - ❖ The same can be achieved by adding transitions in the signal that alert the receiver to the start, middle or end of the pulse.
 - ❖ These points can reset the clock in case the receiver's clock is out of synchronization

Effect of lack of synchronization

- Consider a situation where the receiver has a shorter bit duration
- Let the sender send 10110001, while the receiver receives 110111000011



a. Sent



b. Received

Example

In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 kbps?

How many if the data rate is 1 Mbps?

Solution

At 1 kbps, the receiver receives 1001 bps instead of 1000 bps.

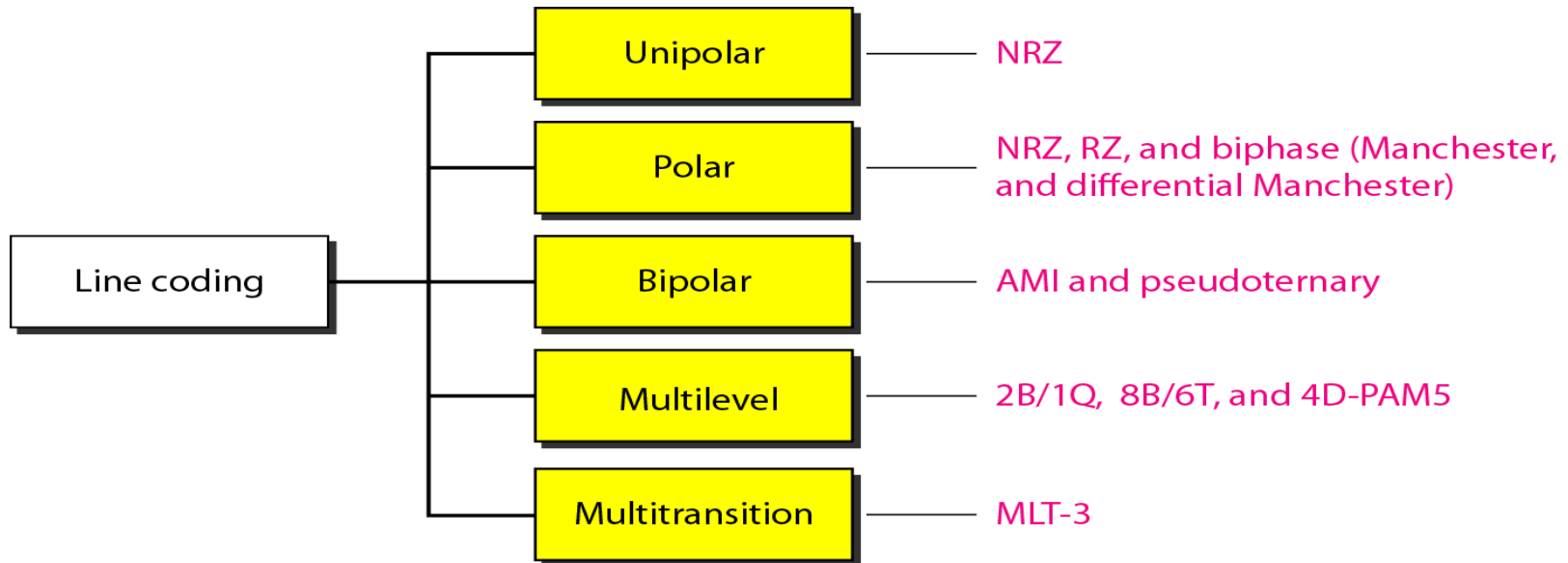
1000 bits sent	1001 bits received	1 extra bps
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At 1 Mbps, the receiver receives 1,001,000 bps instead of 1,000,000 bps.

1,000,000 bits sent	1,001,000 bits received	1000 extra bps
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Digital-To-Digital Conversion

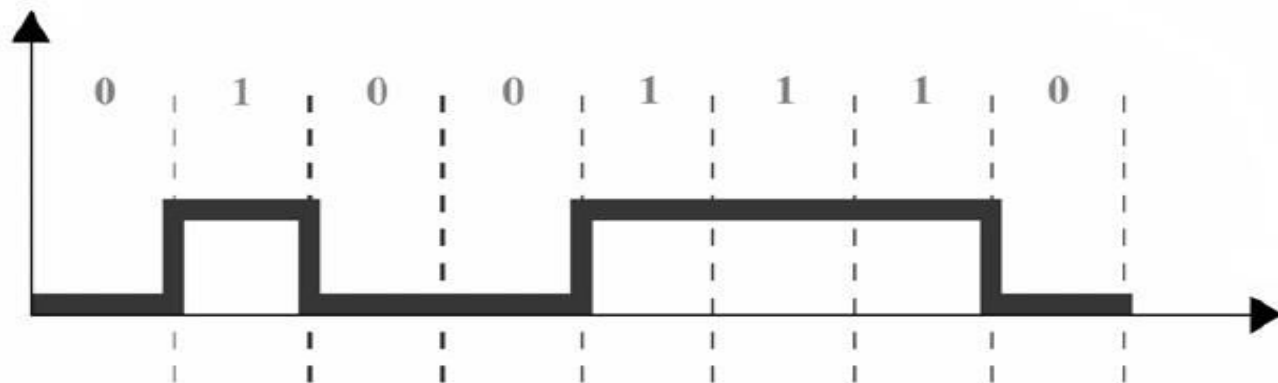
Line Coding Schemes (types of encoding)



Line Coding Schemes: Unipolar

Unipolar: non-return-to-zero (NRZ) [no return in middle]

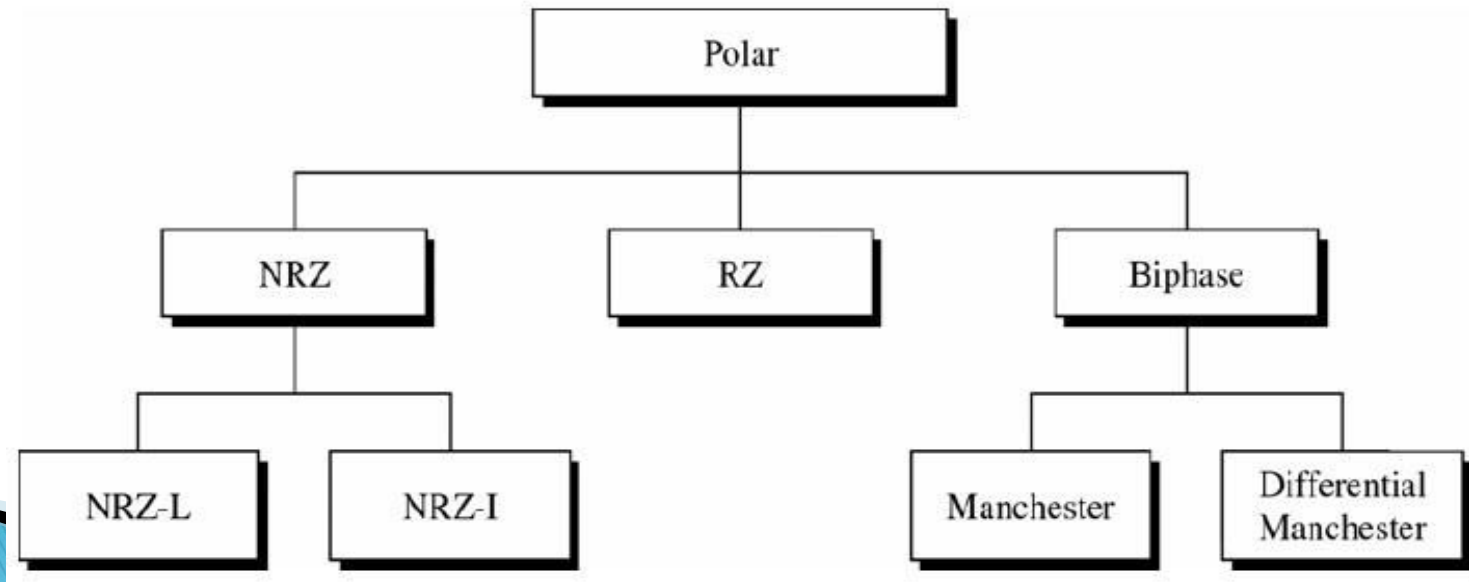
- Encoding is simple , with only one technique in use
- all the signal levels are **on one side of the time axis**, either above or below
- It is called **Unipolar** because it uses only one **polarity**
- positive voltage defines bit 1 and the zero voltage defines bit 0



Line Coding Schemes: Polar

Polar

- Polar encoding uses two voltage levels, positive and negative (on both sides of time axis)
 - ❖ The voltage level for 0 bit can be positive (+ve) and the voltage level for 1 can be negative
- 3 subcategories: NRZ, RZ, Biphase

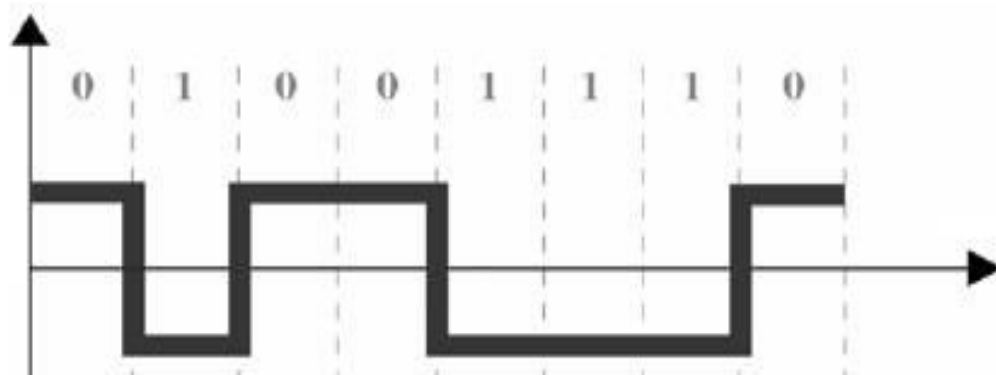


Line Coding Schemes: Polar

Polar: NRZ

Non-return-to-zero-level (NRZ-L)

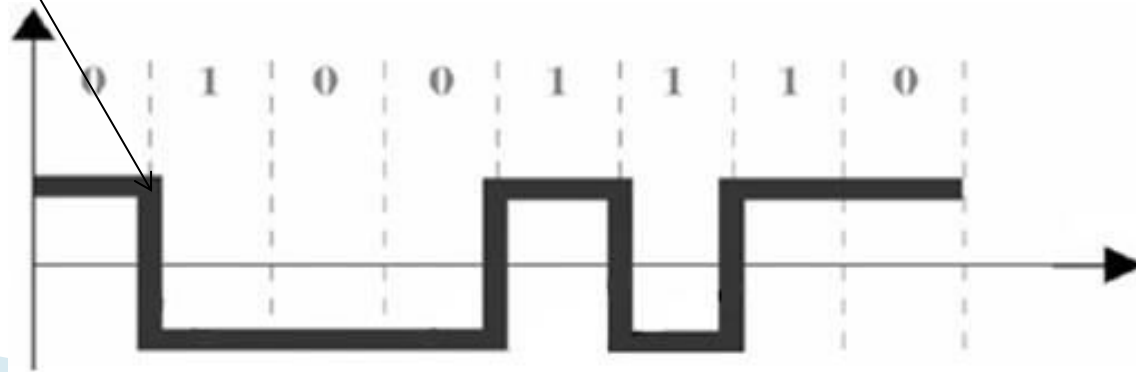
- A +ve voltage means the bit is a 0 and a -ve voltage means the bit is a 1 (vice versa may also be true)



Line Coding Schemes: Polar

Polar: NRZ : Non-return-to-zero-Invert (NRZ-I)

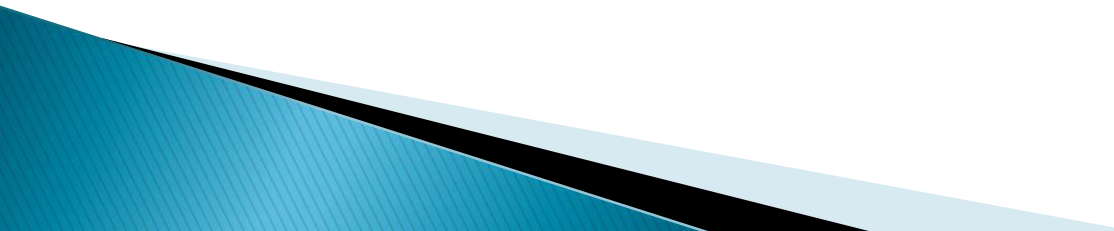
- the change or lack of change in the level of the voltage determines the value of the bit
 - ❖ If there is no change, the bit is 0;
 - ❖ if there is a change, the bit is 1
- A transition (low-to-high or high-to-low) at the beginning of a bit time denotes a binary 1 for that bit time; no transition indicates a binary 0



NRZ-L and NRZ-I both have a DC component problem and baseline wandering, it is worse for NRZ-L. Both have no self synchronization & no error detection. Both are relatively simple to implement.

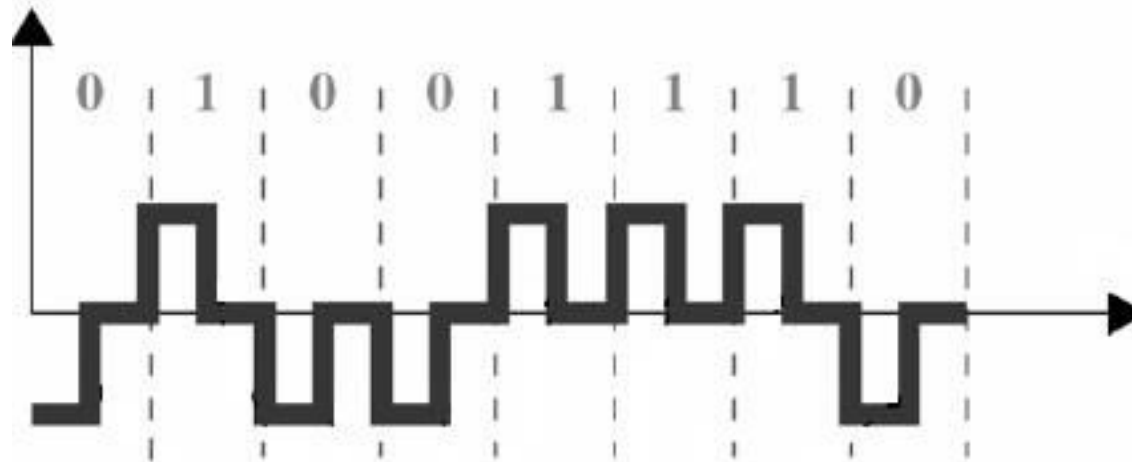
Line Coding Schemes: Polar

Polar: Return-to-zero (RZ)

- Any time, data contains long strings of 1's or 0's, receiver can lose its timing (synchronization problem of NRZ)
 - One solution is **RZ** encoding which uses 3 values; Positive, Negative and Zero
 - Signal changes not between bits but during each bit
- 

Line Coding Schemes

- Here, +ve voltage means 1 and a -ve voltage means 0,
- but unlike NRZ-L, half way through each bit interval, the signal returns to zero
- A 1 bit is represented by positive to zero and a 0 is represented by negative to zero transition



Line Coding Schemes

Problem with RZ

- Main problem with RZ encoding is that it requires two signal changes to encode one bit
 - therefore occupies more bandwidth
 - But it is most effective solution so that receiver cannot lose timing (no DC component problem)
- Complexity: uses three levels of voltage

Line Coding Schemes: Polar

Biphase

- Best existing solution to the problem of synchronization
- **Signal changes at the middle of bit interval but does not stop at zero**
 - Instead it continues to the opposite pole

There are two types of biphase encoding

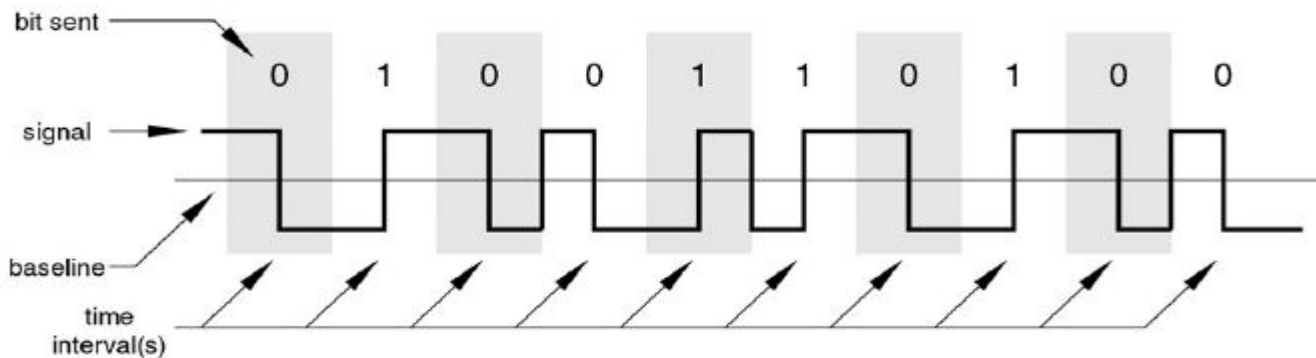
1. **Manchester**
2. **Differential Manchester**

Line Coding Schemes: Polar

Biphase: Manchester

- Transition in middle of each bit period
- Transition serves as clock and data
- Used by IEEE 802.3 (10Mbps Ethernet)
- Negative-to-Positive Transition (low to high) = 1
- Positive-to-Negative Transition (high to low) = 0

Manchester Encoding

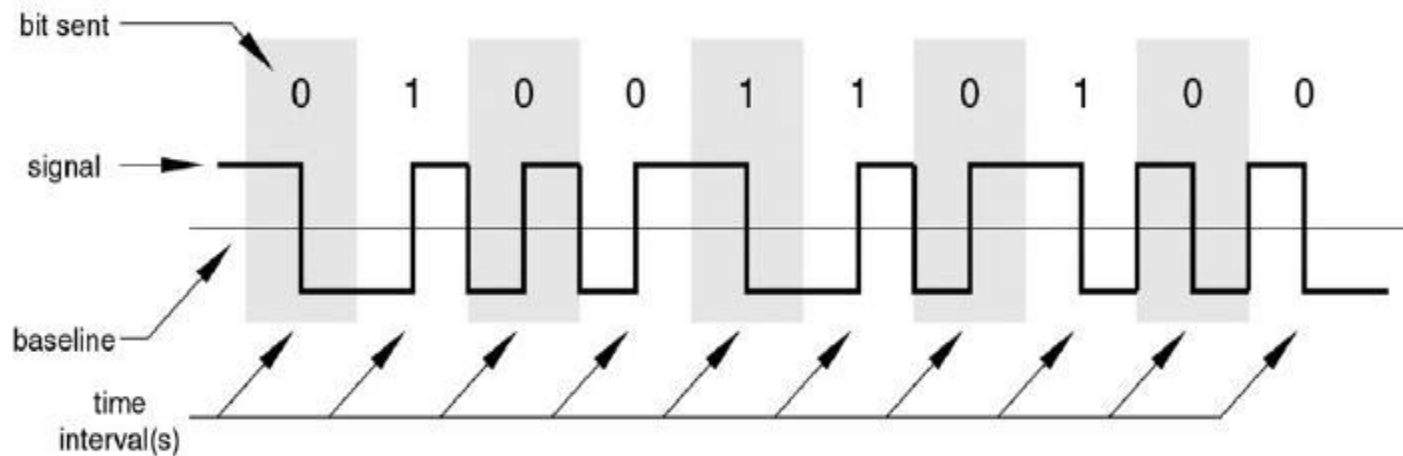


Line Coding Schemes: Polar

Bi-phase: Differential Manchester

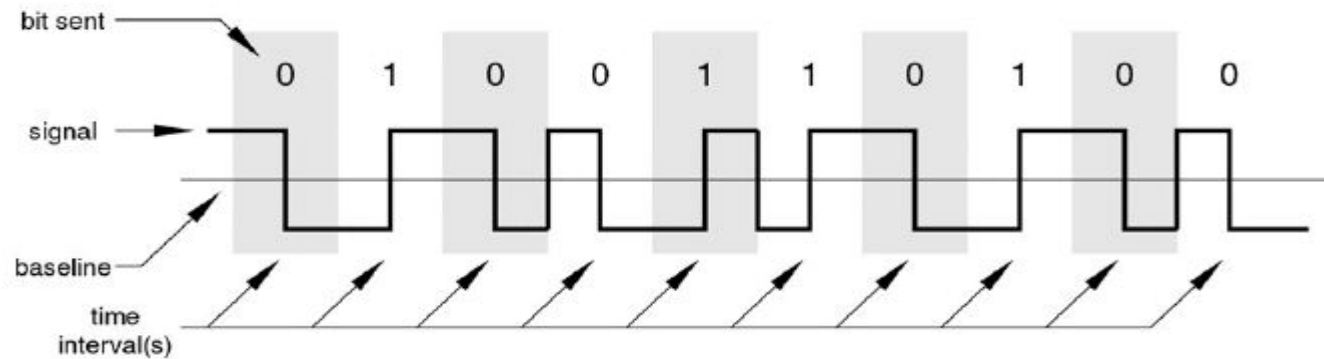
- Mid-bit transition is clocking only (for synchronization)
- Transition at start of a bit period represents zero
- No transition at start of a bit period represents one
- Note: this is a differential encoding scheme
- Used by IEEE 802.5 (Token ring)

Differential Manchester Encoding

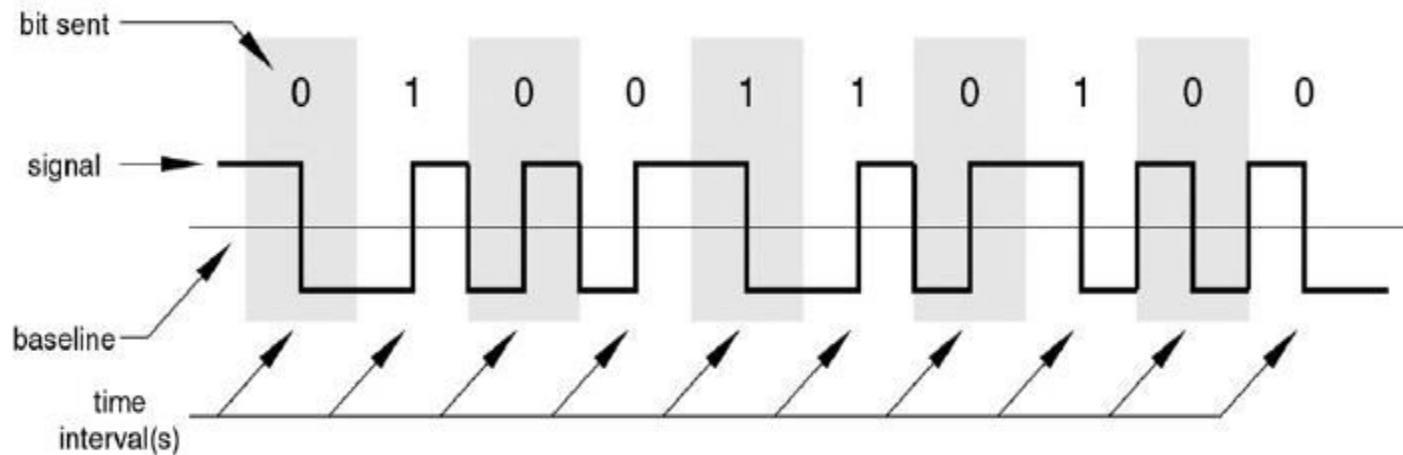


Line Coding Schemes: Polar

Manchester Encoding



Differential Manchester Encoding



Line Coding Schemes: Bipolar

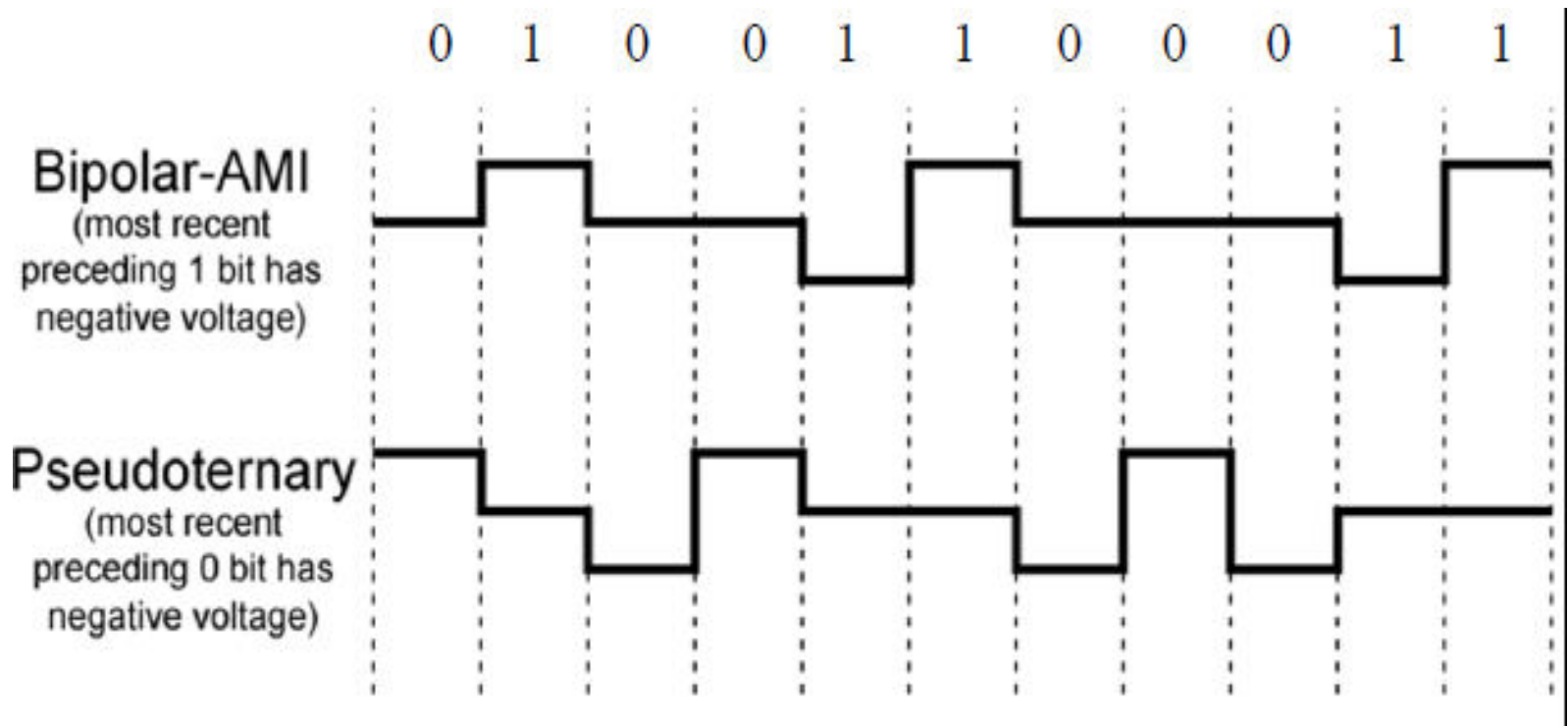
Bipolar encoding

- There are three voltage levels; +ve, -ve, zero to represent the symbols (note not transitions to zero as in RZ)
- Voltage level for one symbol is at “0” and the other alternates between +ve & -ve.
- **Bipolar Alternate Mark Inversion (AMI)**: the “0” symbol is represented by zero voltage and the “1” symbol alternates between +V and -V.
- **Pseudoternary** is the reverse of AMI: the “1” symbol is represented by zero voltage and the “0” symbol alternates between +V and -V

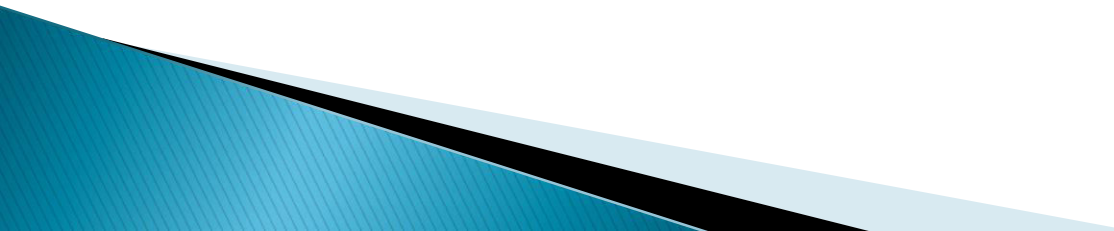
Line Coding Schemes: Bipolar

Bipolar encoding

Bipolar Alternate Mark Inversion (AMI) and Pseudoternary



Alternate Mark Inversion(AMI)

- Pros and Cons:
 - There will be no loss of synchronization if a long string of 1s occurs
 - Each 1 introduces a transition, and the receiver can resynchronize on that transition
 - A long string of 0s would still be a problem
 - No error detection
- 

Digital-To-Digital Conversion

- Representation of digital data by using digital signals
- Techniques for digital-to-digital conversion
 - ❖ Line coding
 - ❖ Block coding
 - ❖ Scrambling

Block Coding

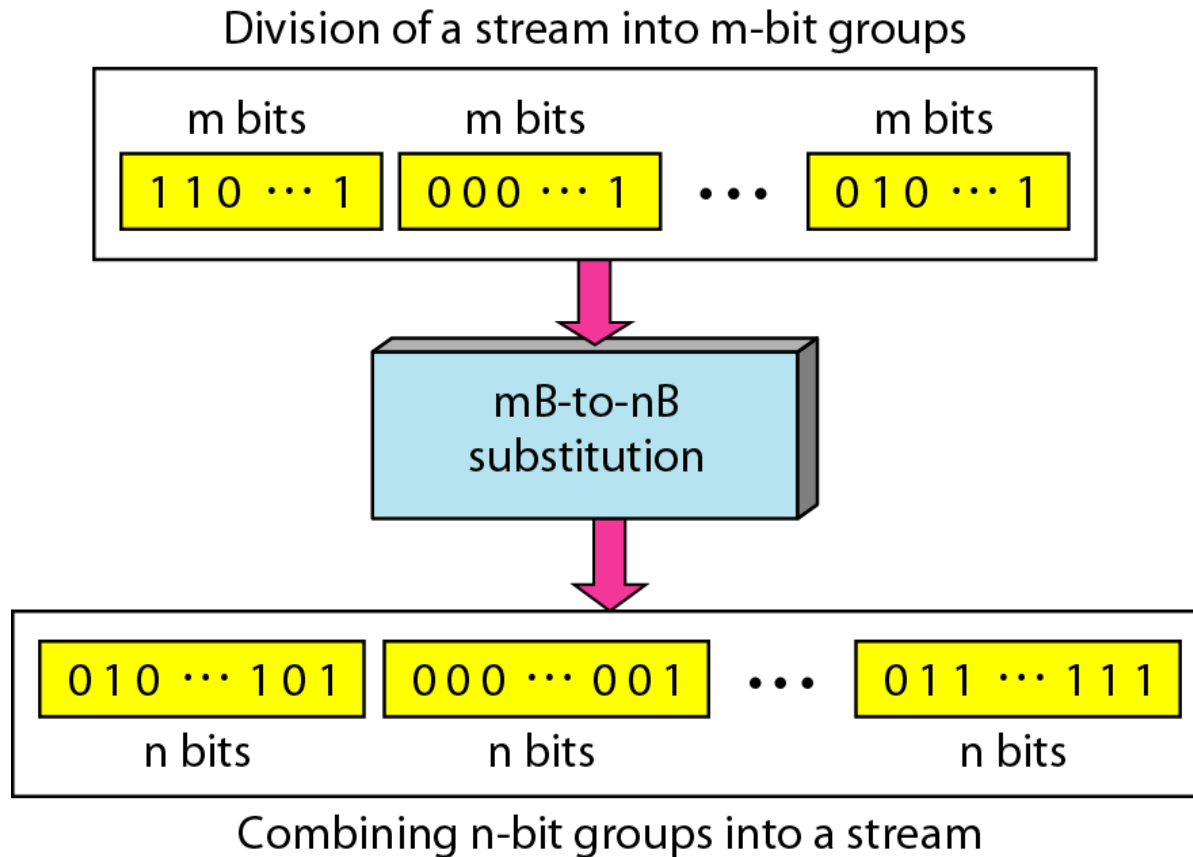
- For a code to be capable of error detection, we need to add redundancy, i.e., extra bits to the data bits.
- **Block coding** is normally referred to as *mB/nB coding*; it replaces each m -bit group with an n -bit group, where n is larger than m
- Block coding is done in **three steps**: *division, substitution and combination*.
- The resulting bit stream prevents certain bit combinations
 - that when used with line encoding would result in DC components or poor sync. quality.

Block Coding Concept

Three steps of block coding

- In the **division step**, a sequence of bits is divided into groups of *m bits*
 - ❖ For example, in 4B/5B encoding, the original bit sequence is divided into 4-bit groups
- In the **substitution step**, we substitute an m-bit group for an n-bit group
 - ❖ For example, in 4B/5B encoding we substitute a 4-bit code for a 5-bit group
- Finally in **combination step**, the n-bit groups are combined together to form a stream
 - The new stream has more bits than the original bits

Block coding concept

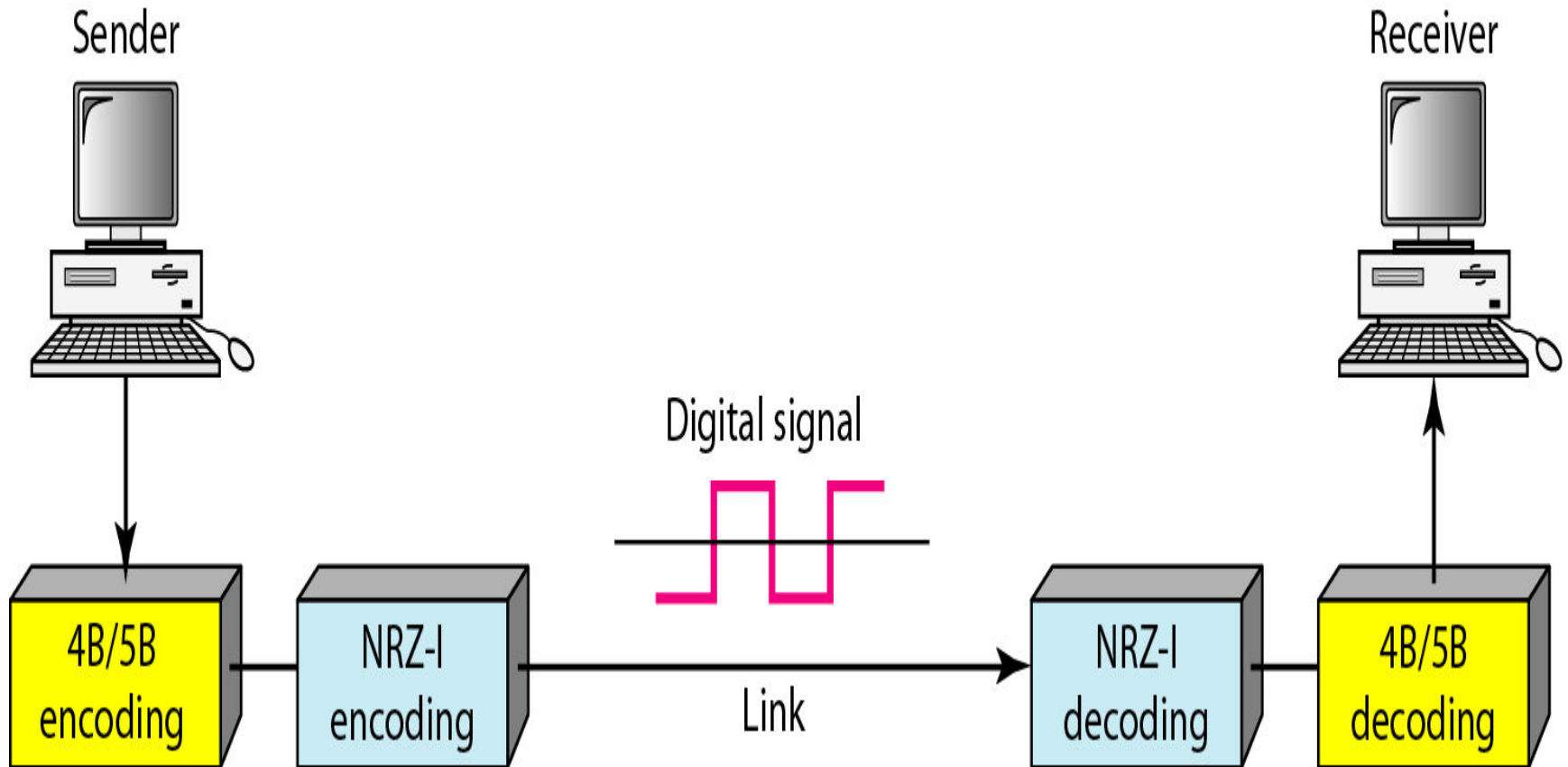


Using block coding 4B/5B with NRZ-I line coding scheme

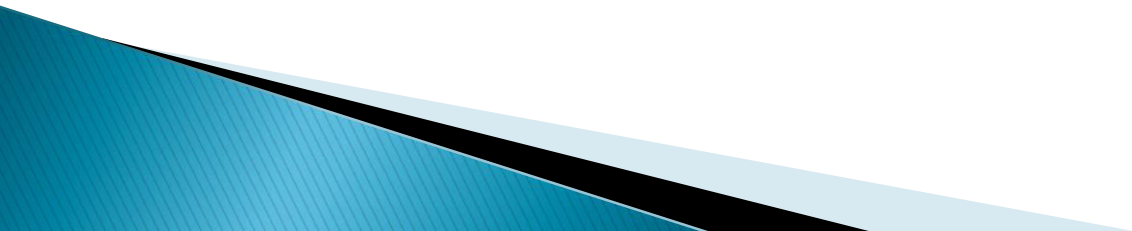
4B/5B coding scheme was designed to be used in combination with NRZ-I

- **NRZ-I** has a good signal rate, but it has a **synchronization problem**
 - A long sequence of 0s can make receiver clock lose synchronization
- **One solution to synchronization issue:** prior to encoding with NRZ-I, change the bit stream to avoid having a long stream of 0s
- The **4B/5B** scheme is the answer to achieve this goal
- Block-coded stream does not have **more than three** consecutive 0s
- **At the receiver**
 - the NRZ-I encoded digital signal is first decoded into a stream of bits and
 - then decoded to remove the redundancy

Using block coding 4B/5B with NRZ-I line coding scheme



4B/5B: Redundancy

- A 4 bit data word can have 2^4 combinations
 - A 5 bit code word can have $2^5=32$ combinations
 - We therefore have $32 - 16 = 16$ unused groups for 4B/5B
 - Some of the extra/unused groups are used for control/signaling purposes.
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4B/5B Mapping codes

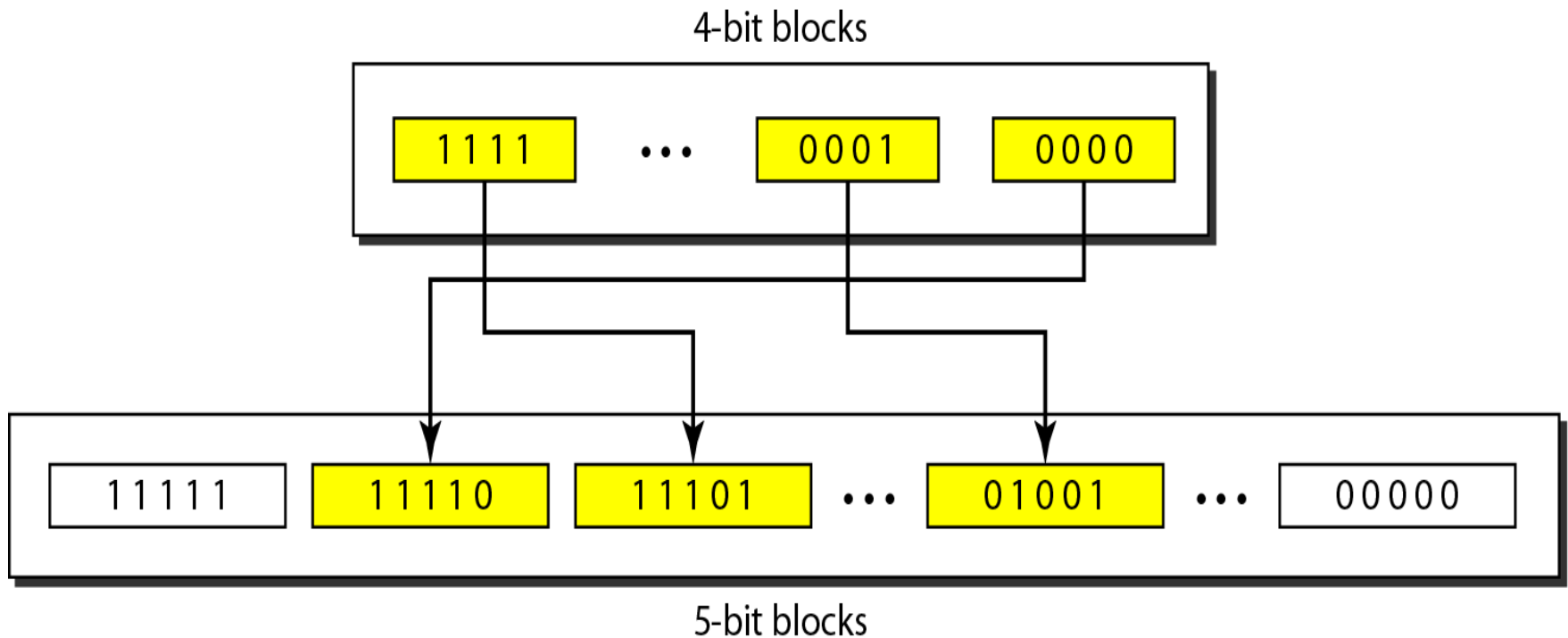
- ▶ The 5-bit codes are selected in such a way that there are no more than one leading 0 (left bit) and no more than two trailing 0s (right bits)
 - Thus, when sent back-to-back, no pair of 5-bit codes results in more than three consecutive 0s being transmitted
- ▶ Thus, when different groups are combined to make a new sequence, there are never more than three consecutive *0s being transmitted*
- ▶ If a 5-bit group arrives that belongs to the unused portion of the table
 - the receiver knows that there is an error in the transmission

The 5 bit words are pre-determined in a dictionary and are chosen in such a way so that there will be at least two transitions per block of bits

4B/5B mapping codes

<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Substitution in 4B/5B block coding



ANALOG-TO-DIGITAL CONVERSION

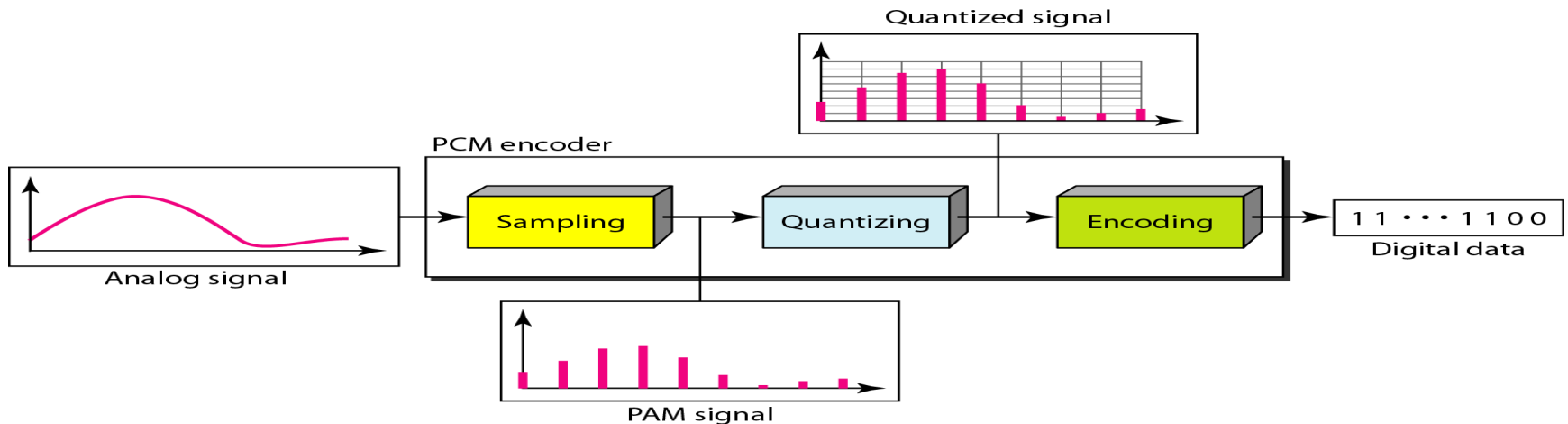
A digital signal is superior to an analog signal because it is more robust to noise and can easily be recovered, corrected and amplified

Two techniques, **pulse code modulation** and **delta modulation**.

PCM

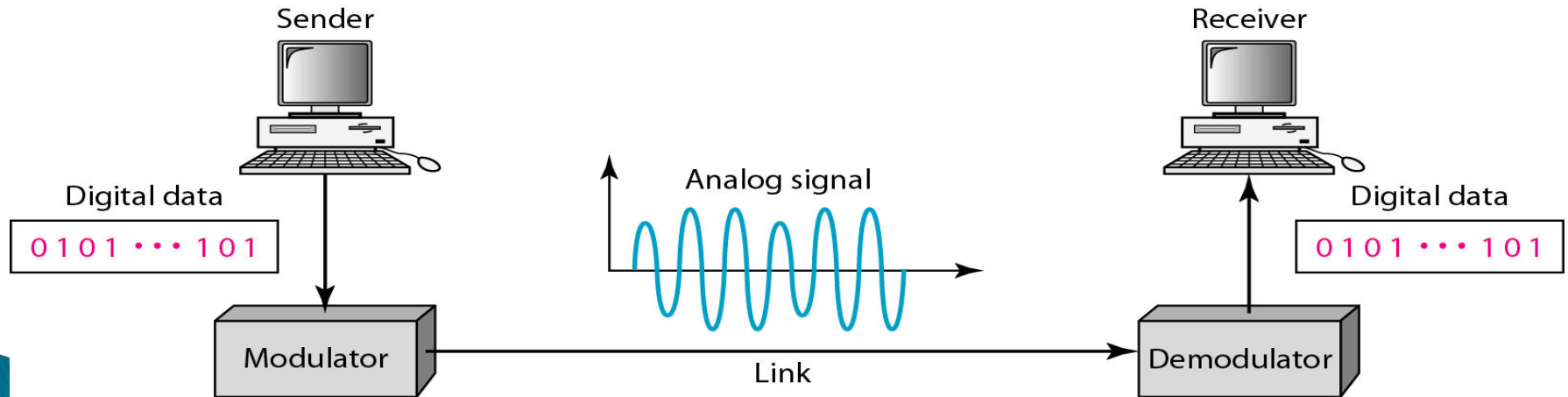
PCM consists of three steps to digitize an analog signal

- Sampling
- Quantization
- Binary encoding

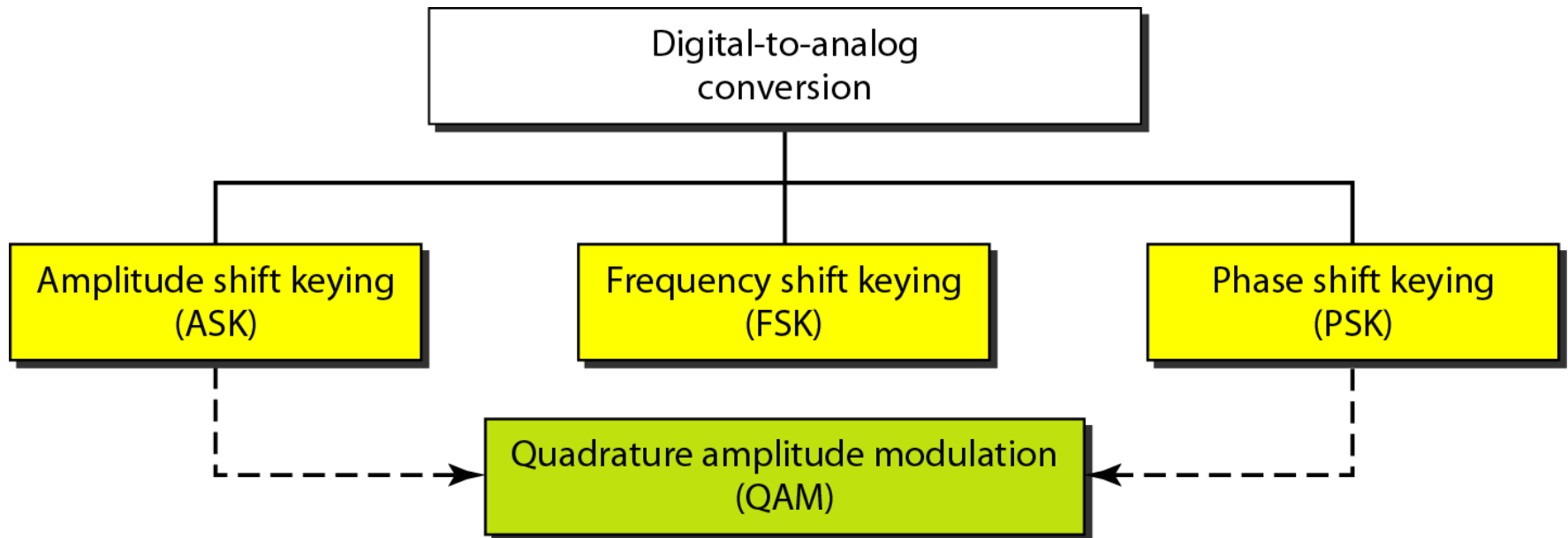


DIGITAL-TO-ANALOG CONVERSION

- **Digital-to-analog** conversion is the process of changing one of the characteristics of digital data to an analog signal based on the information .
- Digital data needs to be carried on an analog signal



Types of digital-to-analog conversion



Analog-to-Analog Conversion

Analog-to-analog conversion is the representation of analog information by an analog signal

- *Types of analog-to-analog modulation*
 - Amplitude Modulation
 - Frequency Modulation
 - Phase Modulation